

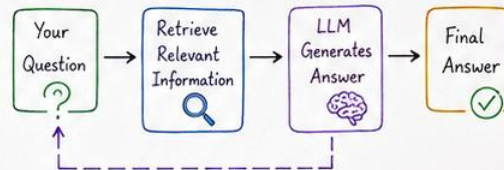


⌋ RAG ⌋

(Retrieval-Augmented Generation)

1. What is RAG?

RAG (Retrieval-Augmented Generation) is an AI framework that combines **information retrieval** from external sources with the power of **Large Language Models (LLMs)** to generate **accurate, context-aware** and **up-to-date** answers.



2. Why RAG?

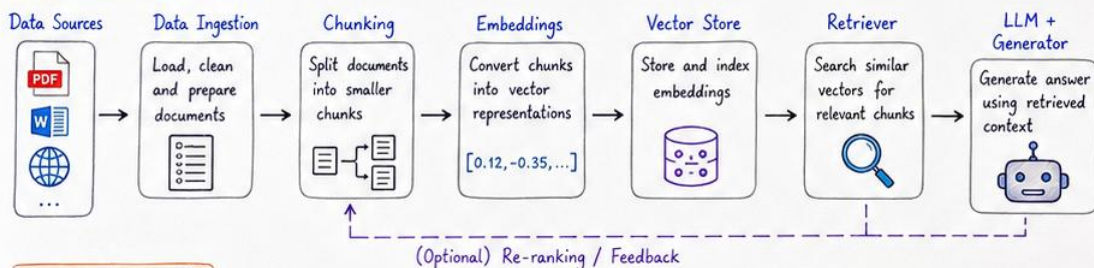
- Reduces **hallucinations**
- Provides **up-to-date** information
- Adds **domain-specific knowledge**
- Improves **factual accuracy**
- Builds **trust** in AI outputs



3. How RAG Works? (Workflow)

- ① User asks a question.
- ② System retrieves relevant documents/passages from a knowledge source.
- ③ Retrieved content is provided to the LLM along with the question.
- ④ LLM generates an answer using both the question and the retrieved context.
- ⑤ Answer is returned to the user.

4. Architecture of RAG



5. Key Components

- **Knowledge Base** : Collection of domain documents.
- **Retriever** : Finds relevant information from the knowledge base.
- **Vector Store** : Stores embeddings for fast similarity search.
- **LLM (Generator)** : Uses retrieved context + question to generate answer.
- **Prompt / Context** : Combines question and retrieved content as input to LLM.



6. Benefits of RAG

- ✓ More accurate and factual answers
- ✓ Real-time and up-to-date knowledge
- ✓ Reduces hallucination
- ✓ Works with private / custom data
- ✓ Transparent (source-able responses)

7. Use Cases



☆ RAG bridges the gap between LLMs and real-world data, enabling smarter, safer and more trustworthy AI applications. ♥

RAG

(Retrieval-Augmented Generation)

Bringing Knowledge to LLMs for Better Answers

8. Advanced RAG Techniques

★ Hybrid Search

Combines keyword (BM25) and semantic (vector) search for better retrieval.



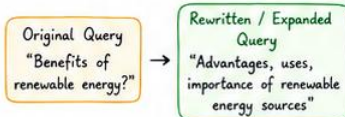
★ Re-ranking

Re-ranks retrieved documents using a cross-encoder or LLM to keep the most relevant ones.



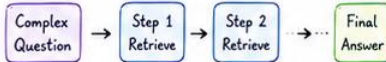
★ Query Transformation

Expands or rewrites the user query to improve retrieval (e.g., using LLM).



★ Multi-step RAG

Breaks complex questions into sub-questions and retrieves information step by step.



9. RAG Variants



Naïve RAG

Retrieve relevant docs and pass them directly to the LLM with the question.



Advanced RAG

Adds re-ranking, better prompts and context formatting for higher accuracy.



Modular RAG

Each step (retrieve, rerank, generate) is a separate module. Easy to upgrade.



Agentic RAG

An agent decides what to retrieve, what tools to use and how to answer iteratively.



Graph RAG

Uses knowledge graphs to retrieve relationships and context more effectively.

10. RAG Prompt / Context Format

System / Instruction

You are a helpful assistant. Use the provided context to answer the question.

Context (Retrieved Information)

Doc 1: ...
Doc 2: ...
Doc 3: ...

} Most relevant information passed to the LLM

Question

(User's Question)

Answer

(Generated by LLM using the context)



11. Evaluation of RAG

Retrieval Evaluation

- Precision@k
- Recall@k
- MRR (Mean Reciprocal Rank)
- NDCG (Normalized Discounted Cumulative Gain)



Generation Evaluation

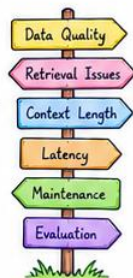
- Faithfulness (Grounded in context)
- Answer Relevance
- Correctness
- Fluency
- Hallucination Rate



★ Good RAG = Relevant retrieval + Faithful generation

12. Challenges of RAG

- ⚠ Quality of source data affects the answers.
- ⚠ Irrelevant or noisy retrieval leads to wrong answers.
- ⚠ Long context may exceed token limits.
- ⚠ Latency increases with multiple steps.
- ⚠ Keeping vector store and data up-to-date.
- ⚠ Evaluating faithfulness is still a hard problem.



13. RAG vs Fine-tuning

Feature	RAG	Fine-tuning
Knowledge Source	External documents (dynamic)	Model parameters (static)
Data Update	Easy (update documents)	Hard (retrain model)
Cost	Lower	Higher
Hallucination	Lower (with good retrieval)	Higher
Transparency	High (sources can be shown)	Low
Customization	Moderate	High

When to use?

- ★ Use RAG when knowledge changes frequently and you need factual, source-backed answers.
- ★ Use Fine-tuning when you need deep behavior or style customization.



RAG empowers LLMs with real-world knowledge, reduces hallucinations, and delivers accurate, trustworthy and up-to-date answers! ♥



RAG

(Retrieval-Augmented Generation)

Smarter Retrieval. Better Context. Reliable Answers.

14. Key Terms to Know

- ★ **Corpus / Knowledge Base:** Collection of all source documents.
- ★ **Passage / Chunk:** Small piece of text retrieved for relevance.
- ★ **Embedding:** Numerical representation of text.
- ★ **Similarity Search:** Finding most similar vectors.
- ★ **Top-k:** Top k most relevant results.
- ★ **Context Window:** Max tokens LLM can take as input.
- ★ **Grounding:** Making the answer based on retrieved facts.



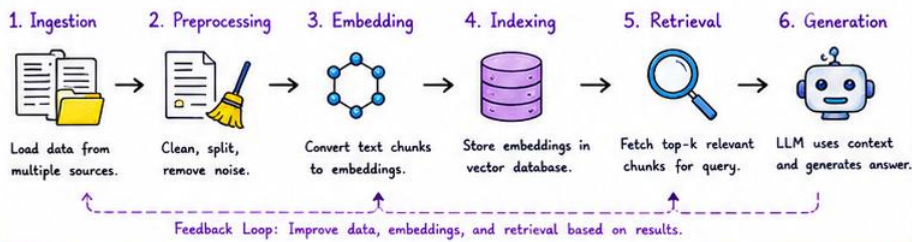
15. Best Practices for Building RAG

- ✓ Use high-quality, clean and well-structured data.
- ✓ Chunk smartly (semantic or small meaningful chunks).
- ✓ Use strong embedding models.
- ✓ Tune Top-k value as per use case.
- ✓ Add metadata/filters for better retrieval.
- ✓ Keep content updated and remove outdated info.
- ✓ Monitor performance and user feedback continuously.



Good Practices
= Better Answers

16. RAG Pipeline in Detail



17. Common Vector Databases

	Pinecone	Managed vector DB, scalable and easy to use.
	Weaviate	Open-source, supports hybrid search and metadata filters.
	Milvus	High-performance, scalable, built for large-scale vectors.
	Chroma	Developer friendly, lightweight and open-source.

18. Evaluation Challenges in RAG

- ⚠ No single "correct" ground truth for open-ended questions.
- ⚠ Retrieval quality and generation quality both matter.
- ⚠ Hard to measure faithfulness automatically.
- ⚠ Different LLMs can give different answers.
- ⚠ Human evaluation is often required.



19. Improving RAG Performance



20. Applications of RAG

 Enterprise Search • Search company documents • Policies, manuals, SOPs	 Customer Support AI • Answer FAQs • Provide accurate product info	 Research Assistant • Find papers • Summarize and answer questions	 Legal AI • Search case laws • Summaries of contracts	 Healthcare AI • Medical knowledge retrieval • Patient info (Q&A)	 Education AI Tutoring • Explain concepts • Provide learning materials
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RAG in Action

Across industries, RAG helps AI systems stay accurate, relevant and trustworthy by connecting LLMs with real-world knowledge.



RAG is the bridge between the power of LLMs and the truth of real-world data.

Retrieve Well. Generate Better. Build Trust. ♥



Image 4

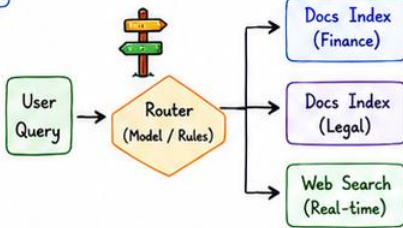
RAG

(Retrieval-Augmented Generation)

★ Building RAG Systems for the Real World ★

1. Query Routing

- Choose the best data source for a query.
- Route to indexes or tools.

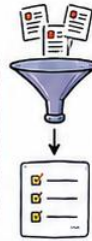


2. Metadata Filtering

- Use metadata to filter documents.
- Returns only relevant and trusted results.

Example Filters

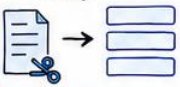
- ✓ Source = "Annual Report"
- ✓ Year >= 2023
- ✓ Department = "HR"
- ✓ Access = "Public"



3. Chunking: Semantic vs Fixed

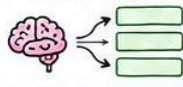
Fixed Chunking

- Split by size (tokens/characters).
- Simple but may cut meaning.



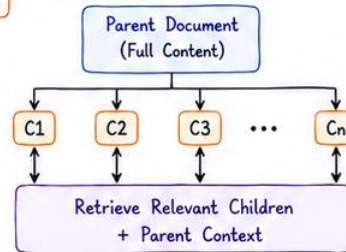
Semantic Chunking

- Split by meaning and context.
- More coherent and relevant chunks.



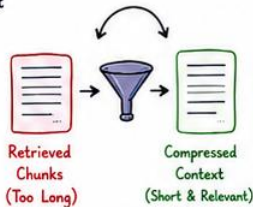
4. Parent-Child Retrieval

- Store large docs as Parent.
- Retrieve small chunks (Children).
- Return relevant chunks + parent context.



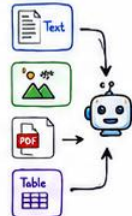
5. Context Compression

- Reduce retrieved context before sending to LLM.
- Techniques:
 - ✓ Summarization
 - ✓ Extractive summary
 - ✓ Relevance pruning



6. Multi-Modal RAG

- Work with text, images, PDFs, tables.
- Useful for reports, invoices, research, dashboards.



7. Knowledge Graph RAG

- Combine vector search with knowledge graphs.
- Capture relationships between entities.



8. Agentic RAG Architecture

- ✓ Plan
- ✓ Decide what to retrieve
- ✓ Use tools
- ✓ Iterate until done

Tools

- Retriever
- Web Search
- Calculator
- Code Executor
- Database



9. RAG Security & Guardrails

- ✓ Data Privacy
- ✓ Access Control
- ✓ Content Safety
- ✓ Hallucination Guard
- ✓ Citations & Transparency

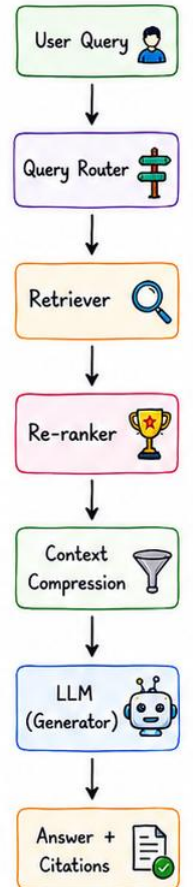


10. Real-world Enterprise RAG Stack

- End-to-end stack for production RAG.
- Scalable, reliable, monitored.
- Deploy at enterprise level.



RAG Pipeline (End-to-End Flow)



Build RAG systems that are accurate, reliable and secure.

Right Information + Right Context + Right Model = Right Answer ♥

